## 12 B. Pyramid of biomass

### Diagnostic Features

- Pyramid of biomass represents the total biomass or standing crop (dry weight) of organisms in each trophic level at a particular time.
- In aquatic ecosystem, the bottom of the pyramid is occupied by the producers, which comprises very small organisms (algae and phytoplanktons) possessing the least biomass and so the value gradually increases towards the tip of the pyramid.
- Therefore, here the pyramid of biomass is always inverted in shape.

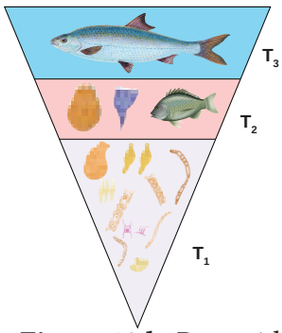



**Figure 12 b:** Pyramid of biomass in aquatic ecosystem

$T_1$ - Producers | $T_2$ - Herbivores | $T_3$ - Secondary consumers |

## 12 C. Pyramid of energy

### Diagnostic Features

- Pyramid of energy represents the number of joules transferred from one trophic level to next.
- The bottom of the pyramid of energy is occupied by the producers. There is a gradual decrease in energy transfer at successive trophic levels from producers to the upper levels.
- Therefore pyramid of energy is always upright.

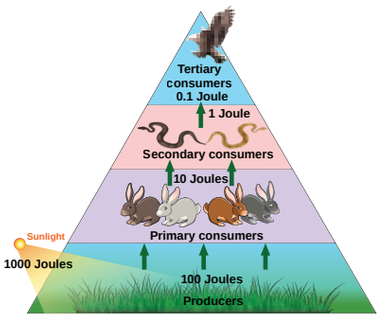



**Figure 12 c:** Pyramid of Energy

# IV - Solving the Problems

## Exercise 13: To verify Mendel's Monohybrid cross

**NOTE:** Student have to work in pairs to perform this experiment and record the data in the observation and record note book with the help of the teacher.

Need not consider this Monohybrid cross experiment for Board Practical Examination.

**Aim:**

To verify Mendel's Monohybrid cross.

**Principle:**

When two pure lines with contrasting traits of a particular character (phenotype) are crossed to produce the next generation ($F_1$ generation), all the members of the progeny are of only one phenotype, i.e. of one of the two parents. The phenotype that appears is called dominant and the one that does not appear is called recessive. When the $F_1$ plants are selfed, the progeny i.e. the $F_2$ generation, is in the ratio of 3 dominant : 1 recessive (¾ : ¼ of 75% : 25%). This reappearance of the recessive phenotype in $F_2$ generation, verifies Mendel's Monohybrid cross.

**Requirements:**

64 yellow and 64 green plastic beads, all of exactly same shape and size (when beads are not available, pea seeds may be painted and used). Plastic beakers, petri dish and a napkin / hand towel.



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